

Experiment 'p_6_6_alternating_sum_seq' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_6_alternating_sum_seq

Natural language statement: null

Method signature: p_6_6_alternating_sum_seq(inputs: seq<int>) returns (alt_sum: int)

Ensures

- alt_sum == sum_array(inputs, 1)

Functional Code Given

```
function sum_array(inputs: seq<int>, i: int): int
{
  if |inputs| == 0 then 0
  else inputs[0] * i + sum_array(inputs[1..], i * -1)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_6_alternating_sum_seq(inputs: seq<int>) returns (alt_sum: int)`

The method should respect the following contract:

`ensures alt_sum == sum_array(inputs, 1)`

The contract uses the following dafny code:

```
function sum_array(inputs: seq<int>, i: int): int
{
  if |inputs| == 0 then 0
  else inputs[0] * i + sum_array(inputs[1..], i * -1)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnOK5opLeMo4RLaRwwvkGWr8nPRKS

Model: gpt-5.5-2026-04-23

Created at: 1780663941

Finish reason: stop

Usage: 408 tokens in, and 6058 tokens out

```
{
  alt_sum := 0;
  var idx := 0;
  var sgn := 1;

  assert inputs[0..] == inputs;
  while idx < |inputs|
    invariant 0 <= idx <= |inputs|
    invariant alt_sum + sum_array(inputs[idx..], sgn) == sum_array(inputs, 1)
    decreases |inputs| - idx
  {
```

```

var oldIdx := idx;
var oldSgn := sgn;
var oldAlt := alt_sum;

assert oldIdx < |inputs|;
assert |inputs[oldIdx..]| > 0;
assert (inputs[oldIdx..])[0] == inputs[oldIdx];
assert (inputs[oldIdx..])[1..] == inputs[oldIdx + 1..];
assert sum_array(inputs[oldIdx..], oldSgn) == inputs[oldIdx] * oldSgn + sum_array(
    ↪ inputs[oldIdx + 1..], oldSgn * -1);

alt_sum := oldAlt + inputs[oldIdx] * oldSgn;
sgn := oldSgn * -1;
idx := oldIdx + 1;

assert alt_sum + sum_array(inputs[idx..], sgn) == sum_array(inputs, 1);
}

assert idx == |inputs|;
assert inputs[idx..] == [];
assert sum_array(inputs[idx..], sgn) == 0;
assert alt_sum == sum_array(inputs, 1);
}

```

Final Program

```

function sum_array(inputs: seq<int>, i: int): int
{
  if |inputs| == 0 then 0
  else inputs[0] * i + sum_array(inputs[1..], i * -1)
}

method p_6_6_alternating_sum_seq(inputs: seq<int>) returns (alt_sum: int)
  ensures alt_sum == sum_array(inputs, 1)
{
  alt_sum := 0;
  var idx := 0;
  var sgn := 1;

  assert inputs[0..] == inputs;
  while idx < |inputs|
    invariant 0 <= idx <= |inputs|
    invariant alt_sum + sum_array(inputs[idx..], sgn) == sum_array(inputs, 1)
    decreases |inputs| - idx
  {
    var oldIdx := idx;
    var oldSgn := sgn;
    var oldAlt := alt_sum;

    assert oldIdx < |inputs|;

```

```

    assert |inputs[oldIdx..]| > 0;
    assert (inputs[oldIdx..])[0] == inputs[oldIdx];
    assert (inputs[oldIdx..])[1..] == inputs[oldIdx + 1..];
    assert sum_array(inputs[oldIdx..], oldSgn) == inputs[oldIdx] * oldSgn + sum_array(
        ↪ inputs[oldIdx + 1..], oldSgn * -1);

    alt_sum := oldAlt + inputs[oldIdx] * oldSgn;
    sgn := oldSgn * -1;
    idx := oldIdx + 1;

    assert alt_sum + sum_array(inputs[idx..], sgn) == sum_array(inputs, 1);
}

assert idx == |inputs|;
assert inputs[idx..] == [];
assert sum_array(inputs[idx..], sgn) == 0;
assert alt_sum == sum_array(inputs, 1);
}

```

Total Token Usage

Input tokens: 408

Output tokens: 6058

Reasoning tokens: 5629

Sum of 'total tokens': 6466

Experiment Timings

Overall Experiment started at 1780663940723, ended at 1780664091713, lasting 150990ms (150.99 seconds)

Iteration #1 started at 1780663940724, ended at 1780664091713, lasting 150989ms (150.99 seconds)